

MODELING ROBOTS

Fundamentals of Robotics — Lecture 02

Why modelling

- Purpose of robotics: endow a mechanical machine with some degree of intelligence.
- A robot's intelligence is expressed through the physical interaction with the surrounding world; thus it is crucial to understand what a robot can do with its own body.
- As humans, we have the same problem: we learn our body and how to use it via our experiences since the first days of our lives.

Learning vs mathematical model

- In some cases, using experiences and learning to figure out how to transfer intent into action is possible and useful.
- However, most of the times we know quite a bit of the physical structure of a robot; this knowledge comes from physics and is expressed by a **mathematical model**.
- Mathematical models can be manipulated by mathematical means, with guaranteed results for **control** and **motion planning**.

What to model

- The math of robotic modelling: capture **geometry**, **kinematics** and **dynamics** using a quite general mathematical formalism.
- What we need to understand: how to create and use a model in **simulation** and **design**.

Taxonomy and terminology

Manipulators robots with a fixed base.

Mobile robots robots with a mobile base.

Robotic manipulators

A robotic manipulator consists of:

- **Arm** — ensuring mobility.
- **Wrist** — located at the end of the arm, enabling dextrous operation.
- **End-effector** — executing the robot's tasks.

From the geometric point of view, the manipulator is a sequence of rigid bodies (**links**), connected by mechanical articulations (**joints**).

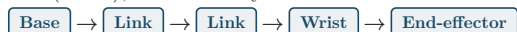


Fig. 1: Manipulator structure: base, rigid links connected by joints, wrist and end-effector.

Joints

- Joints provide the structure with its necessary mobility.
- Two types:
 - **Prismatic joints**: enable a relative translational motion between the links.
 - **Revolute joints**: enable a relative rotational motion between the links.



Fig. 2: Joint types: prismatic (relative translational motion) and revolute (relative rotational motion).

Definitions

Dexterity A robot's ability to cope with a variety of objects and actions: how robots can interact and handle objects and take the necessary actions on the objects.

Stiffness Ability of a body to resist deformation. In formal terms, the amount of force required to induce a motion along a DoF.

Degrees of freedom and workspace

- In a mechanical structure a **degree of freedom** (DoF) defines a specific mode in which the robot can move.
- Typically each joint is endowed with an actuator (e.g., a brushless, or a linear motor) and provides the structure with one degree of freedom.
- The more the degrees of freedom, the more flexible the machine.
- **Workspace**: the area that the end effector can reach; its structure very much depends on the number and on the type of the joints.

- Typical rotation axes: 1) waist rotation, 2) shoulder rotation, 3) elbow rotation, 4) wrist rotation, 5) gripper rotation.

Types of manipulators

According to the structure of the different DoF: **Cartesian**, **Cylindrical**, **Spherical**, **SCARA**, **Anthropomorphic**.

Cartesian	three prismatic joints with three mutually orthogonal axes; workspace is a parallelepiped; good mechanical stiffness and accuracy everywhere in its workspace; the exclusive presence of prismatic joints reduces the structure's dexterity.
Cylindrical	one of the joints is replaced by a revolute joint; wrist accuracy decreases with horizontal stroke; good stiffness; workspace is a portion of cylinder; primarily used to move heavy loads within cylindrical cavities.
Spherical	two prismatic joints are replaced with revolute joints; workspace is a portion of a hollow sphere; reduced accuracy when the radial stroke increases; primarily used for machining.
SCARA	Selective Compliance Assembly Robot Arm: two revolute joints and one prismatic joint; axes of motion are parallel; high stiffness to vertical loads, compliance to horizontal loads; well suited to vertical assembly tasks and manipulation of small objects; positioning accuracy decreases with the distance from the first axis.
Anthropomorphic	three revolute joints; the axis of the second and third joint are orthogonal to the axis of the first; evident similarity with the human arm (second joint = shoulder, third = elbow); the three revolute joints maximise dexterity; wrist position accuracy is different in the workspace; by far the most widespread in industry (59% of the installation).

Wrist

- All different manipulators have an end-effector attached to a wrist.
- The end-effector is usually a gripper, but can be a different thing according to the task (e.g., a welding gun, a spray gun, a screwdriver).
- The configuration of the wrist that maximises dexterity is spherical (three revolute joints).
- Each joint of a wrist provides an additional DoF.

Mobile robots

A mobile robot is characterised by a mobile base. Two types:

- **Wheeled robots**: a rigid body (chassis) and a system of wheels that provide motion with respect to the ground; can be connected to a system of trailers connected by revolute joints.
- **Legged robots**: multiple rigid bodies connected through revolute (and sometimes) prismatic joints; some of the links form lower limbs and feet that stay in contact with the ground and provide locomotion. Not considered in this course.

Wheel types

Wheeled robots use three types of wheels:

- **Fixed**: can only rotate about an axis passing through the centre and orthogonal to the wheel plane.
- **Steerable**: has two axes of rotation: the standard axis typical of all wheels, and a vertical axis passing through the centre.
- **Caster wheel**: similar to a steerable wheel, but its vertical axis has an offset from the centre of the wheel; the wheel swivels around when bending and automatically aligns itself with the direction of motion of the chassis.

Note By combining different types of wheels we can obtain different types of kinematic structures.

Drive configurations

Differential drive

- Two actuated fixed wheels with the same axis; a caster wheel to keep the robot statically balanced.
- The robot can rotate by applying a different velocity to the wheels, or move straight applying the same angular velocity.
- It can rotate on the spot by setting the two speeds to opposite values.

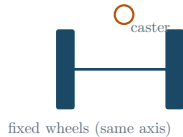


Fig. 3: Differential drive: two actuated fixed wheels on a common axis plus a caster wheel.

Synchro drive

- Three steerable wheels connected by a chain.
- Two motors: one rotates the three wheels “synchronously”, the other transmits the motion (synchronously) to all of them.
- It can do the same kinematics of the differential drive, but it needs only one motor to move straight.

Tricycle

- Two wheels that are fixed and actuated by a motor.
- The third wheel is steerable and its rotation around the vertical axis is governed by another motor.
- It is also possible that the two motors operate both on the steering wheel to turn it and to secure locomotion.

Carlike

- Very similar to a tricycle except that it has two turning wheels.
- Traction can be on the fixed wheels (rear-wheel traction) or on the front wheel (front-wheel traction).
- Contrary to a differential drive robot, a car-like (and a tricycle) cannot turn on the spot and has a limited curvature radius.

Ackerman steering

- Back in 1758 Erasmus Darwin was driving his horse-drawn carriage; the two front wheels turned of the same angle and one of the wheels (the external one) slipped sideways, causing the carriage to tip over its driver.
- The solution was devised by Lankensperger in Munich and patented by his English agent (Ackerman) in London: the two wheels have to follow the same circular trajectory.
- In order to follow the same circle the internal wheel has to turn more than the external one; this is solved, in modern cars, by the geometry of the axle.
- The Ackerman steering can be generalised to the case of trailers.



Fig. 4: Ackerman steering: the two front wheels follow the same circular trajectory (the internal wheel turns more than the external one).

Motion constraints

- Contrary to manipulators, mobile robots do not usually have a limited workspace: they can reach any location in the Euclidean space.
- However, they do have motion constraints:
 - a differential drive robot is quite flexible, but it cannot move sideways along the axis connecting the actuated wheels: **nonholonomic constraint**;
 - a carlike has a limited curvature radius.

Exception: omnibot

- An exception is an omnibot.
- It is characterised by three autonomously actuated caster-wheels (usually in symmetric positions).
- It can move in any direction.

Additional types of robots

- Mounting an anthropomorphic manipulator and a differential drive mobile base improves the robot’s working abilities.
- Dexterity can be improved by additional degrees of freedom (7 for the Kuka LWR).

More sophisticated end-effectors

- A very active research area has been on the development of sophisticated anthropomorphic hands, usable for:

- rehabilitation and assistive purposes (e.g., as a replacement for severed human hands);
- creation of sophisticated anthropomorphic robots.

Humanoid robots

- Putting together several types of robots we can create humanoid robots, but this is out of the scope of this course.

Modelling: kinematics

Kinematics of manipulators

- **Kinematics**: the motion of a robotic structure with respect to a generic frame; for a manipulator, given the joint position, find the position and orientation of the end effector.
- **Differential kinematics**: given the joint positions and motion (and their velocity), find the velocity of the end effector.
- **Forward kinematics**: find the end effector motion as a function of the joint motion.
- **Inverse kinematics**: find the joint motion as a function of the “desired” end-effector motion and configuration.

Kinematics of mobile robots

- The robot’s kinematics requires a correct understanding of the motion constraints.
- The kinematic model translates the instantaneous motion of the motors into an instantaneous motion of the robot as a whole, which accounts for the robot’s current configuration and for its motion constraints.